

JARED DUONG

Hamilton, Ontario

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Education

University of Guelph

September 2024 – April 2028

Bachelor of Engineering in Computer Engineering (Co-op)

Guelph, ON

Relevant Coursework: Digital Systems Design (VHDL), Computer Architecture, Electric Circuits, Data Structures, Discrete Structures, Numerical Methods, Applied Differential Equations, Probability & Statistics

Technical Skills

Programming: C, C++, Python, Java, VHDL, SQL, C#, HTML

Embedded / Hardware: Arduino, FPGA (Nexys-3), Digital Logic Design, Sensor Integration, Soldering

Tools: Git, Vivado, ModelSim, MATLAB, IntelliJ, VS Code

CAD: SolidWorks, Fusion 360, AutoCAD

Experience

Production Engineering Intern

May 2025 – June 2025

Hartmann Canada

Brantford, ON

- Analyzed PLC sensor data and machine cycle logs to identify bottlenecks, reducing downtime by **15%**.
- Improved production monitoring logic, increasing output consistency by **10%**.
- Investigated recurring mechanical and process failures and implemented corrective actions to improve system reliability.
- Collaborated with operators and engineers to reduce manufacturing waste and improve line efficiency.

Game Developer

April 2023 – June 2023

McMaster University

Hamilton, ON

- Developed interactive educational games using **C#** and Unity to improve student engagement.
- Built responsive web interfaces using **HTML, CSS, and JavaScript**, supporting over **200** users across devices.

Projects

16-Bit ALU Design and FPGA Implementation | VHDL, FPGA, Digital Logic

January 2026 – February 2026

- Designed modular 1-bit ALU slices and composed a scalable 16-bit ripple-carry architecture in **VHDL**.
- Implemented arithmetic and logic operations including add, subtract, **AND**, **OR**, **XOR**, and **NOT**.
- Developed a **VHDL** testbench to verify functionality across multiple input cases.
- Validated behavior on **Nexys-3 FPGA** hardware using simulation and debugging tools.

4x4x4 LED Cube with Custom PCB | Arduino, PCB Design, Digital Logic

Mar 2025 – Apr 2025

- Designed and soldered a 64-LED cube using discrete logic ICs (**74HC04**, **SN74HC08**) and a custom-routed **PCB**.
- Programmed multiplexed LED control patterns using an **Arduino Nano** in **C/C++**.
- Implemented layered scanning logic to control 3D LED addressing and timing.
- Designed and 3D-printed a structural base to house power and control electronics.

Automated Arduino Wheelchair Prototype Project | Arduino, C++, SolidWorks

Sept 2024 – Dec 2024

- Developed an Arduino-powered autonomous wheelchair prototype integrating sensors and motor control, improving navigation accuracy by **25%**.
- Performed center-of-mass and tipping analysis to redesign mechanical components, reducing tipping risk by **15%**.
- Conducted cost evaluation and optimization, lowering prototype expenses by **10%** without sacrificing performance.

Leadership / Extracurricular

Robotics Club – Mechanical Team Member

Sept 2024 – Present

University of Guelph

Guelph, ON

- Designed and programmed autonomous robotic systems using **C++** and **Arduino** for the Canadian International Robotics Competition.
- Optimized sensor integration and control algorithms to improve system responsiveness and performance.
- Designed and prototyped mechanical components using **CAD**, collaborating with teammates to debug and refine designs.